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Hybrid gender agreement in Polish

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Chapter 1

Theoretical background

1.1 Previous work on hybrid agreement

Agreement can be found between a word with some inherent features (a controller) and another, related word (a target). We can see attributive agrrement between a demonstrative and a noun in (1), where the noun is plural, so the demonstrative has to be *these* to agree with the number of the noun. In (2) there is predicate agreement: the subject is the singular pronoun *she*, so the verb has to agree with the number and take the form *reads*. These are examples of agreement marking a syntactic dependency, but it can also be used to mark a discourse dependency. This can be found in (3), where the personal pronoun *she* has to be feminine to have the same referent as the earlier noun phrase *little sister*.

(1) **These dogs** belong to my neighbor.

(2) **She reads** every morning.

(3) I have a little sister, **she**'s 5 years old.

There are cases where the justification for an agreement is rather semantic than syntactic. This can be found in (4): the copular *are* has to be plural, despite the nouns in the subject noun phrase being singular – this is because there is a group reading of the coordinate phrase and there is plural agreement to show that.

(4) John and Mary **are** rich.

Hybrid agreement is possible when there are agreement alternatives, but the controller remains the same. Corbett (2023) provides a simple example of that in English: both sentences in (5) are acceptable. The controller of agreement here is the word *family*, the same in both sentences, and there are two agreement alternatives: *have* in (5a) and *has* in (5b). One of them is available, which is known from the lexical semantics of the noun: it can refer to a group of people. The other alternative comes from the morphosyntactic specification: the noun is singular.

(5) a. The family have lost everything.
b. The family has lost everything.

This is an example of hybrid number agreement, but same is possible for any other feature that requires agreement in a given language. This study focuses on gender agreement in Polish. Before the system of genders in Polish is described, this section describes Corbett's work on agreement, as it is the main basis for the study conducted later.

Corbett proposed the Agreement Hierarchy, which orders different types of agreement. This hierarchy is used to formulate the constraint visible in (6).

attributive > predicate > relative pronoun > personal pronoun

Figure 1.1: The Agreement Hierarchy (Corbett, 1979)

(6) *Constraint of the Agreement Hierarchy (to be revised)*
For any controller that permits alternative agreements, as we move rightwards along the Agreement Hierarchy, the likelihood of agreement with greater semantic justification will increase monotonically (that is, with no intervening decrease).

We can see this on the example of the word *family*. Starting with the leftmost element of the Agreement Hierarchy, (7a) shows that singular (syntactic) agreement is acceptable, but (7b) shows that plural (semantic) is not. However, with the next step of the hierarchy, predicate agreement, both options are possible, as (5) has shown. Plural agreement is also available with a personal pronoun, which can be found in (8).

(7) a. This family
b. * These family

(8) I saw my family yesterday. They seemed to be tired.

This constraint was created for the core instances of hybrid agreement, such as the one shown above (a lexical hybrid), and constructional mismatches, an example of which can be seen in (9).

(9) The girl and the boy have been on the playground.

Both nouns in the coordination are singular, but the number in the auxiliary verb *have* agrees with the plural number suggested by the whole construction, instead of the number of the nouns within the construction.

The problem with the Agreement Hierarchy constraint in the form shown above is that it is based on "semantic justification", when hybrid agreement can occur without one of the agreement options having a clear semantic justification. This is clear when we expand the plane of possible hybrid agreements and look at what else can trigger agreement.

Alternative sources of agreement information are visible on the left of Table 1.2. The table also gives an idea of where hybrid agreement comes from: it happens whenever different sources give conflicting information about the value of a given feature. This is represented by the thick line which separates the situations where the source suggests syntactic agreement (above the line) from the situations where it suggests semantic agreement (below the line).

		controller type			
		split hybrids	lexical hybrids	constructional mismatch	extraneous
Hierarchy of Agreement Sources	cell specification (1)				
	morphosyntactic specification (2)				
	lexical semantics (3)				
	complex nominal constructions (4)				
	extraneous source (5)				

Figure 1.2: Core (in gray) and generalized semantic agreement.

This figure includes two more examples of hybrid agreement: the split hybrid *ręka* (meaning 'hand') in Polish and the Norwegian *pancake* sentences with extraneous agreement information.

Polish number system used to have three possible values: singular, dual and plural. Dual is not used anymore, but the word *ręka* (meaning 'hand') is one of few where the relics of dual number can be found: the form used historically as locative dual can now be used as locative singular. In modern Polish the form morphologically resembles masculine or neuter, while the noun itself is feminine. This leads to a situation, where attributive agreement with the locative singular form of the noun can be masculine, as shown in (10). [1]

(10) w lewym ręku
in left-sg.loc.M hand-sg.loc
in the left hand

There are two conficting sources of agreement: the morphosyntactic specification of the noun is feminine, but the specification of the (singular locative cell) can be

[1] In this example the M1, M2 and M3 forms are syncretic; whenever that is the case, the M label will be used.

masculine/neuter. The question then is: which of the options can be considered "semantic" agreement? One could say the feminine is semantic, because it makes it possible to track the antecedent later in the discourse, but this explanation can be considered to stretch the notion of "semantic" agreement to cover a less-than-semantic example.

On the other hand there is Norwegian "default agreement", that is triggered when there is no straightforward agreement controller, for instance when a sentence contains logical metonymy. As an example Corbett uses the sentence visible below.

(11) Magnus Carlsen og Bobby Fischer hadde vær-t **super-t**.
Magnus Carlsen and Bobby Fischer had be-pst.ptcp **super-sg.n**
'Magnus Carlsen and Bobby Fischer would have been wonderful.'

Here, by *Magnus Carlsen and Bobby Fischer* speaker means *having Magnus Carlsen and Bobby Fischer play chess.* Instead of agreeing with the features suggested by the coordination, there is extraneous (default) agreement. Here the conflicting sources of agreement are: coordination of proper nouns that refer to two men and the metonymy, which triggers the "default agreement". Here the example is considered to be more than semantic and therefore not fitting in the core semantic agreement.

Including split hybrids and examples with extraneous agreement information in the analysis forces us to expand beyond semantic agreement and consider agreement options that are less or more than semantic. This is what Corbett did by proposing generalized semantic agreement, along with the revised constraint of Agreement Hierarchy.

(12) *Revised constraint of the Agreement Hierarchy*
For any controller that permits alternative agreements, as we move rightwards along the Agreement Hierarchy, the likelihood of agreement with a greater degree of generalized semantic justification will increase monotonically (that is, with no intervening decrease).

The aim of the current paper is to see whether the Polish gender agreement pattern really is compatible with the Agreement Hierarchy and with the Hierarchy of Agreement Sources. Polish has a rich gender system, there is gender agreement with adjectives, some verbs and pronouns and it has large lexical and syntactic resources available. First, it is necessary to understand the system of grammatical gender in Polish.

1.2 Grammatical gender in Polish

There are multiple options of defining grammatical genders in Polish. The one chosen for this study was proposed by Mańczak (1956), where he differentiates between 5 genders: masculine personal, masculine animate, masculine inanimate, feminine and neuter. The difference between the genders is the easiest to observed when looking at appropriate forms of adjectives in the accusative case. Adjectives agree their gender with nouns and the accusative case provides the most diversity in terms of gender forms.

The example presented in Table 1.3 comes from Mańczak (1956). There are 5 nouns and one adjective for all of them used as an example: *mężczyzna* (meaning 'man'), *pies* (meaning 'dog'), *stół* (meaning 'table'), *kobieta* (meaning 'woman'), *dziecko* (meaning 'child') and *dobry* (meaning 'good').

<table>
<tr><th colspan="2">singular</th><th colspan="2">plural</th><th rowspan="2">gender of the noun</th></tr>
<tr><th>adjective</th><th>noun</th><th>adjective</th><th>noun</th></tr>
<tr><td rowspan="2">dobrego</td><td>mężczyznę</td><td>dobrych</td><td>mężczyzn</td><td>masculine personal</td></tr>
<tr><td>psa</td><td rowspan="4">dobre</td><td>psy</td><td>masculine animate</td></tr>
<tr><td>dobry</td><td>stół</td><td>stoły</td><td>masculine inanimate</td></tr>
<tr><td>dobrą</td><td>kobietę</td><td>kobiety</td><td>feminine</td></tr>
<tr><td>dobre</td><td>dziecko</td><td>dzieci</td><td>neuter</td></tr>
</table>

Figure 1.3: Examples of nouns in accusative with 5 grammatical genders in Polish with their agreeing adjectives.

The table shows that each of the nouns agrees with a different set of adjective forms in the accusative singular and plural: *-ego* and *-ych* form for masculine personal, *-ego* and *-e* for masculine animate, *-y* and *-e* for masculine inanimate, *-ą* and *-e* for feminine and *-e* in both numbers for neuter. There are 5 possible sets of agreements, which gives 5 possible grammatical genders.

Saloni (1976) proposed an even more elaborate system, where he talked about 9 genders (instead of one neuter there were two and additionally there were three genders for plurale tantum nouns). Such a system is, however, not necessary for the current analysis, therefore for the sake of simplicity we will adopt the 5 genders proposed by Mańczak. For ease of reference we will use for them the names that Saloni used, which are: M1 (masculine personal), M2 (masculine animate), M3 (masculine inanimate), F (feminine) and N (neuter).

1.3 Polish nouns with possible hybrid gender agreement

Corbett (2023) provides the example of a split hybrid *ręka* (meaning 'hand') in Polish, though unfortunately that example cannot be explored further with the resources available for this study. The circumstances under which the noun has developed to its current use are rare and there are no other nouns which can be analyzed in addition to this one. There are, however, three other categories of nouns which can be used to study hybrid gender agreement and these are described in the following subsections. This work focuses agreement with nouns and does not address complex nominal constructions, such as coordination.

1.3.1 Socially gendered nouns

Here the socially gendered nouns are defined as the nouns that refer to people and semantically involve the gender or sex of the referent. Corbett (2023) provides the German *das Mädchen* (meaning 'girl') as an example of lexical hybrids – the grammatical gender of the noun is neuter, but the semantics of the word include the feminine gender. The nouns in this group are similar to the German example. There is a number of words in Polish where the gender of the referent is incongruent with the grammatical gender of the noun, which can lead to hybrid agreement. An example, very similar to the German one, can be found in (13).

(13)	To	dziewczę	jest	niemożliwe!	Ona	w	ogóle	się	nie	uczy!
	this-N	girl-N	is	impossible-N	she	in	general	REFL	not	study

'This girl is impossible! She never studies!

Here the lexical hybrid is essentially the same as the German *das Mädchen* – a noun meaning 'girl' has neuter grammatical gender. In the example we find three words agreeing with the noun: demonstrative *to* agreeing with the grammatical gender, predicate adjective *niemożliwe* agreeing with the grammatical gender and the personal pronoun *ona* in the second sentence agreeing with the semantic gender.

The socially gendered nouns are an instance of lexical hybrids: the conflicting sources of agreement are between the lexical semantics and the morphosyntactic specification of the noun. Most of those nouns refer primarily to women or girls in an expressive (often derogative) manner, but there are different combinations of conflicting values of gender. The process of choosing the nouns for this category and the combinations of conflicting genders are described in the Methodology chapter.

1.3.2 Incongruent profession nouns

In Polish some nouns referring to people holding certain job titles and functions or practising certain professions have masculine gender and there are no feminine nouns with the same meanings. These nouns are likely to be understood to only refer to men, but for a lack of better term they can be also used to refer to women. This can be seen in the examples (14) and (15).

(14)	Nowa	psychiatra	przepisała	mi	nowe	leki.
	new-F	psychiatrist	prescribed-F	me	new	medicine
(15)	Nowy	psychiatra	przepisał	mi	nowe	leki.
	new-M	psychiatrist	prescribed-M	me	new	medicine

The same noun *psychiatra* is used in both sentences and the only way to tell whether the referent is female or not is by looking at the agreement: feminine in (14) and masculine in (15).

Another option to mark the referent as female is to precede the masculine noun by the word *pani*, which can be translated as 'lady' or 'madam'. It is however possible to omit it.

With recent initiatives to ensure that women are linguistically represented in all fields, there is more discussion about those masculine nouns, specifically for which of them it is difficult to create the feminine counterpart and why is it so (Szpyra-Kozłowska, 2019; Szpyra-Kozłowska & Laidler, 2022). The current study takes advantage of this area of linguistic research by using some examples of those masculine nouns as opportunities to find hybrid gender agreement.

1.3.3 Depreciative forms of M1 nouns

In his paper Saloni points out that for all M1 nouns in plural nominative and vocative case it is possible to create an alternative form, which he calls "depreciative" (Saloni, 1988). This form is created very systematically, specifically by inflecting the noun the same way a noun of a different gender would be inflected. The M1 noun *profesor* (meaning 'professor') will serve as an example here. *Profesor* represents the nominative singular form of the noun. There are two options for the nominative plural form of this noun: *profesorzy* and *profesorowie*, both of these are non-depreciative. To create the depreciative form, it is necessary to look for a noun of a different gender with the same ending. For this example we can use the M3 noun *procesor* (meaning 'processor'), which in plural nominative takes the form *procesory*. Therefore the depreciative form of the noun *profesor* is *profesory*.[2] The example (16) shows the depreciative form used in a sentence, where both attributive agreements and the predicative agreement is non-M1, therefore agreeing with the gender in the specification of the cell.

(16) Te stare profesory przyszły w zabłoconych
these-M2/M3/N/F old-M2/M3/N/F professor-M1 came-M2/M3/N/F in muddy
butach.
shoes.

These old professors came in with muddy shoes.

Despite the name, the purpose of using a depreciative form is not necessarily to deprecate the referent, although it has been found to be a common function of the form. Based on a corpus study, Siuda (2010) reports that the purpose of 66% depreciative forms is to express negative feelings towards the referent, 24% of the forms are used as a stylistic device, 6% are supposed to reduce the distance to the referent and 4% have a different function.

Depreciative forms can be created only for M1 nouns, but they are created by inflecting them as if they were of any other gender. The conflicting sources of agreement are the following: the morphosyntactic specification of the nominative (and vocative) plural is M2, M3, N or F, while the morphosyntactic specification of the noun is M1. This makes depreciatives the only form used in this study that represents a different controller type: a split hybrid.

[2]In plural nominative and vocative forms are syncretic, therefore the same goes for the vocative form.

1.4 Hypothesis

The aim of this study is to investigate the extent to which Corbett's constraints of the agreement hierachy hold on Polish data with hybrid gender agreement. The hypothesis is the following: Polish examples of hybrid gender agreement (socially gendered nouns, profession nouns and depreciative nouns) fit the simple constraint of the agreement hierarchy: there is a monotonous shift from one value of gender to another when going from attributive, through predicate to relative pronoun agreement for all kinds of nouns tested. More semantic agreement is expected in predicate agreement than in attributive and even more in relative pronoun agreement. Furthermore, according to the Hierarchy of Agreement Sources, there should be more semantic agreement with the lexical hybrids (so the socially gendered and profession nouns) than with the split hybrids (depreciative forms).

Chapter 2

Corpus study

2.1 Methodology

2.1.1 Agreement sources

The corpus used for this study is the 300M subcorpus of the National Corpus of Polish (Przepiórkowski, 2012), which was parsed using Trankit (Nguyen, Lai, Veyseh, & Nguyen, 2021). The dependency trees (conforming to the Universal Dependency standard, henceforth UD) obtained this way were searched for words, which could trigger hybrid agreement: profession nouns, socially gendered nouns and depreciative forms of M1 nouns. The dependencies of those words then made it possible to search for attributive agreements (adjectives), predicate agreements (verbs agreeing with their subjects) and relative pronoun agreement. A tagged corpus could have been sufficient for attributive and predicate agreement, but using a parsed corpus gave a better chance of finding the correct relations and agreements within them and made it possible to include relative pronouns in the analysis. The subsections below describe how I chose the nouns for analysis.

2.1.1.1 Profession nouns

Nouns in this category came from two studies about feminitives in Polish. The first was a survey reported by Szpyra-Kozłowska (2019), where participants were asked to create feminitives from 50 masculine nouns, that in modern Polish do not have feminine counterparts at all or those counterparts are rarely used. For each noun participants had the option to give no answer. The second study was an experiment conducted by Szpyra-Kozłowska and Laidler (2022), where they tested whether certain feminitives are less accessible by asking participants to try to pronounce them. They chose 20 feminitives, which are formed from masculine nouns that have consonant clusters such as /kt/ (e.g. *architekt*, meaning 'architect'), /tr/ (e.g. *geriatra*, meaning 'geriatrician') or /rg/ (e.g. *chirurg*, meaning 'surgeon'). According to Klemensiewicz (1957) adding the -ka suffix, the function of which can be to create the feminitve, makes the word difficult to pronounce. Those two studies give a list of 60 nouns in total, for which it is said to be difficult to form feminine counterparts. It is therefore likely that, if there was a female referent of a noun from this list, the masculine noun would be used, leading to hybrid agreement.

Both of the studies determine in some way which of the tested nouns have less available feminitives than others: in Szpyra-Kozłowska and Laidler (2022) those are the nouns for which there are the highest error rates in pronunciation, whereas in Szpyra-Kozłowska (2019) – the nouns for which the most people refused to propose a feminitive. Taking this approach has, however, two downsides: first, it requires an arbitrary threshold that separates the nouns that have an easily formed feminitive from the nouns for which creating the feminine counterpart is more difficult. Second, this approach significantly limits the number of nouns for analysis. Considering this, the current study used all of the nouns from the two studies on Polish feminitves.

For ease of reference, I call all of the nouns in this category *profession* nouns, despite not all of them actually referring to professions. Meanings of all nouns are presented in Table 2.1. An example of semantic agreement with a profession noun can be seen in (17).

(17) Ojciec obecne-j prezydent zastał zastrzelony.
father current-F president.M1 was shot
"Father of the current president was shot."

2.1.1.2 Socially gendered nouns

I chose the nouns for this category using the Polish Academy of Sciences Great Dictionary of Polish available online.[1] With the advanced search option, I searched for nouns in the thematic category HUMAN AS A PHYSICAL BEING: HUMAN PHYSICALITY: SEX. This search yielded 586 results, which then I filtered manually to find the nouns that only refer to humans (and not, for example, abstract nouns or nouns referring to body parts) and that had incongruent grammatical and semantic gender. There were 19 nouns selected this way: 7 of them were grammatically masculine animate and referred to women (e.g. *babsztyl*, meaning 'woman' in a pejorative way), 1 was grammatically masculine inanimate and referred to women (*pasztet*, referring to a woman whom the speaker finds ugly or unattractive), 7 were grammatically neuter and referred to women (e.g. *pannisko*, meaning 'a young woman' in an expressive way), 2 were grammatically neuter and referred to men (e.g. *chłopaczysko*, meaning 'a tall or big young man') and 2 were grammatically feminine and referred to men (e.g. *ciota*, meaning 'a homosexual man' in a pejorative way). An example of semantic agreement with a socially gendered noun can be found in (18) and the full list of nouns in this category with their translations is in Table 2.2.

(18) Ten ciota go straszy policją?
this.M1 faggot.F him threatens police
This faggot threatens him with the police?

[1] https://wsjp.pl/

Noun in Polish	Translation or definition in English	Noun in Polish	Translation or definition in English
adiunkt	someone who holds a PhD title and is employed at a scientific institution	architekt	architect
burmistrz	mayor	chirurg	surgeon
demiurg	demiurge	dramaturg	playwright
drań	scoundrel	dupek	asshole
dureń	fool	dziekan	dean
ekspert	expert	elekt	person elected in a specified post, but has not yet entered office
foniatra	phoniatrician	ftyzjatra	pulmonologist
geometra	geometer	geriatra	geriatrician
ginekolog	gynecologist	głupek	fool
herszt	ringleader	inżynier	engineer
jaskiniowiec	caveman	jeniec	captive
jeździec	rider	kierowca	driver
ksiądz	priest	lump	tramp
magister	master (holder of a university degree)	majster	foreman
metalurg	metallurgist	minister	minister
murarz	bricklayer	muzykolog	musicologist
mędrzec	wise man	naukowiec	scientist
nieuk	dunce	nurek	diver
pajac	clown	pastor	pastor
pediatra	pediatrician	petent	applicant
polityk	politician	porucznik	lieutenant
powstaniec	insurgent	prefekt	prefect
premier	prime minister	prezydent	president
proboszcz	parish priest	psychiatra	psychiatrist
pułkownik	colonel	rektor	person in charge of a university
sierżant	sargeant	starosta	highest administrative officer of a second-level unit of local government, mayor
subiekt	salesman	szklarz	glazier
szpieg	spy	szubrawiec	rascal
upiór	phantom	widz	audience member
wójt	highest administrative officer of a basic-level unit of local government, mayor	zbieg	fugitive

Table 2.1: Profession nouns used in the study, with their translations or definitions in English.

Noun in Polish	Translation or definition in English
babochłop	a woman that resembles a man with her physicality, looks and behavior
babol	a woman that looks or behaves in a way that causes aversion in the speaker
babon	a woman, about whom the speaker feels negatively
babsko	a woman, about whom the speaker feels negatively
babsztyl	a woman, about whom the speaker feels negatively
babus	a woman, about whom the speaker feels negatively
chłopaczysko	a tall or big young man
chłopobaba	a man that resembles a woman with his physicality, looks and behavior (can also mean a woman that resembles a man)
ciota	a homosexual man (can also refer to a woman about whom the speaker feels negatively)
dziewczątko	a girl, whom the speaker likes or to whom they feel compassion
dziewczę	a young girl
dziewuszysko	a girl who causes aversion or contempt in the speaker
kobiecisko	expressively about a woman, augumentative, or to refer to a big woman
kobieciątko	expressively about a woman, diminutive
kobieton	a woman that resembles a man with her physicality, looks and behavior
pacholę	a boy
pannisko	expressively about a young woman, augumentative
pasztet	an ugly and unattractive woman
wamp	a confident, misterious, provocative and attractive woman who seduces men

Table 2.2: Socially gendered nouns used in the study, with their translations or definitions in English.

2.1.1.3 Depreciative forms

For this category no list of nouns was created, since the corpus was annotated and I could search for the depreciative forms by checking the words' morphological description. It is possible to create depreciative forms of surnames, however these were excluded from the analysis, because there was no way of determining the referents and therefore no way of ensuring that there is hybrid agreement. Surname *Machała* found in the corpus can serve as an example: let us say that the *Machała* family bought a new car. Assuming the family consists of at least one man, the speaker can use both (19a) and (19b) to convey that information, depending on their attitude they have towards the situation: (19a) if the speaker is neutral or positive and (19b) if they are negative. However, if there are only women in the family, the speaker can only use (19b). Without the context it is unclear whether *Machały* in (19b) is used as a depreciative or simply to refer to women and therefore we cannot decide if this is an example of hybrid agreement.

(19) a. Machał-owie kupi-li nowe auto.
Machała-pl.nom.m1.ndepr buy-3pl.m1.past new car

b. Machał-y kupi-ły nowe auto.
Machała-pl.nom.m1.depr buy-3pl.f.past new car
"Machałas bought a new car."

Morphological analyzer Morfeusz (Kieraś & Woliński, 2017) can mark forms as surnames which made it easy to exclude them from the analysis. The rest of the forms marked as depreciative was kept, which gave 254 different lexemes being included as depreciative forms. An example of semantic agreement with a depreciative form can be found in ().

(20) Chłopy ze wsi pomaga-li tutaj.
peasants.M1 from countryside helped-M1 here
"Peasants from the countryside helped here."

2.1.2 Extracting agreement occurrences from the corpus

The code used for extracting corpus data in this study can be found in its dedicated GitHub repository.[2] The general idea of how the code works is presented in this section.

2.1.2.1 Finding the agreement controllers

After choosing the nouns to be searched in the corpus, it was necessary to find all of their forms and the forms' morphological descriptions, which I did using the morphological analyzer Morfeusz (Kieraś & Woliński, 2017). First I searched the corpus for occurrences of nouns, which could be controllers of hybrid agreement. Those nouns were saved if they were in the list of gendered or profession noun forms or if they were marked as depreciative in their morphological description. To ensure that no homonyms were accidentaly saved for later analysis, the morphological description of the found word was compared with the one given by Morfeusz. This way, for example, the word *premiera* was excluded if it was an instance of the lexeme *premiera* (meaning 'premiere') in the singular nominative form, but not if it was an instance of the lexeme *premier* (meaning 'prime minister') in the singular accusative form.

[2] https://github.com/bmagdab/hybrid-agreement

Once the right noun was found, I searched the dependency tree for words that can agree their gender with that of the noun. This was the head of the noun (which could be an example of predicate agreement), adjectives attached to the noun (which could be examples of attributive agreement), and relative pronouns referring to the noun. Based on the Universal Dependencies (UD) annotation guidelines, I searched all dependents of the word for a dependency label with a subtype :relcl, which in the UD annotation scheme marks the head of the relative clause. Then I looked for the dependent of the head of the relative clause, that had a UD feature PronType=Rel, which had to be the relative pronoun.[3]

Each occurrence was additionally annotated for the distance between the controller and potential target of agreement (measured in intervening words) and the ordering: whether the target of agreement was the first or not. Once I found the possible agreement targets, I could use the data to look for occurrences of hybrid agreement.

2.1.2.2 Extracting and listing agreement occurrences

To analyse hybrid gender agreement in the found examples, I marked the alternative to the grammatical gender of each agreement controller. For all profession nouns that alternative gender was f, because the only chance of hybrid agreement for those nouns is when the gender of the referent is a woman. For socially gendered nouns the alternative gender was assigned depending on the lexeme and for depreciative forms the alternative gender was always m1, because that is the gender of the lexemes that have depreciative forms.

Once I determined the alternative gender, I excluded plural nouns for which the gender discrepancy was between feminine and neuter. This is because for all types of agreement tested here, the plural feminine and plural neuter forms of targets of agreement are syncretic, therefore there is no hybrid agreement in those cases. This happens, for instance, in (21), where in singular ((21a) and (21b)) there is a difference between the feminine and neuter forms of the word *młoda*. In plural (21c), however, the form is the same for both neuter and feminine, so hybrid agreement cannot be observed. Examples like this were therefore excluded.

(21) a. młod-e dziewczę
young-sg.n.nom girl.sg.n.nom

b. młod-a dziewczę
young-sg.f.nom girl.sg.n.nom

c. młod-e dziewczę-ta
young-pl.f/n/m2/m3.nom girl-pl.n.nom
"young girl"

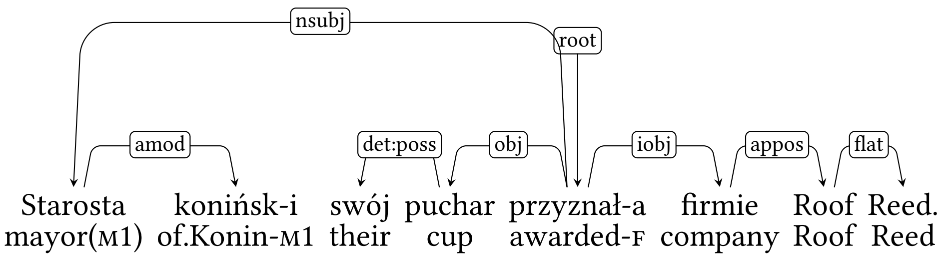
The next step was to find occurrences of different types of agreement. The predicate agreement of a given noun was found in its head on three conditions: first, dependency label between the noun and the head had to be nsubj, which is the UD label for nominal

[3] https://universaldependencies.org/workgroups/newdoc/relative_clauses.html

subjects. Second, the head had to be a verb that inflects for gender, which is only true for verbs annotated as past participles,[4] active and passive participles. Third, the noun had to be in the nominative case. This condition helps filter out situations where a numeral phrase is in subject position, because then the verb does not agree with the subject – it is in 3rd person, singular and neuter form, regardless of the noun. Therefore, predicate agreement cannot be observed in those situations at all and they were excluded.

The search was simpler with attributive and relative pronoun agreement. The only requirement for attributive agreement was that the word was an adjective, which was verified by looking for the tag ADJ in the morphological description of the dependent. As for relative pronouns, all of the ones found earlier were included in the analysis, unless they did not inflect for gender (which was the case, for example, for the word *gdzie* (meaning 'where')).

In (2.1) we can see a sentence from the National Corpus of Polish, which serves as an example of the process of extracting agreement described in this chapter. The word, which can trigger hybrid agreement is *starosta* (meaning 'mayor'). There are two words in the sentence that agree with this word in terms of gender: *koniński* (meaning 'of Konin') and *przyznała* (meaning 'awarded'). All of the information saved for each instance of agreement is shown in Table 2.2.



"The mayor of Konin awarded their cup to the company Roof Reed."

Figure 2.1: Dependency tree for the example sentence.

[4]The annotation scheme used for NKJP uses the past participle label not only for past participles, but also for parts of past tense and conditional mood forms (Przepiórkowski, 2012).

		Occurrence 1	Occurrence 2
type of text		typ_publ	
source of agreement	form	starosta	
	lexeme	starosta	
	morphological description	subst:sg:nom:m1	
	grammatical gender	m1	
	other gender	f	
agreement target	form	przyznała	koniński
	morphological description	praet:sg:f:perf	adj:sg:nom:m1:pos
	gender	f	m1
	distance	3	0
	target first	0	0
	semantic agreement	1	0
type of agreement		predicate	attributive

Figure 2.2: All of the information extracted from the sentence in (2.1)

2.1.2.3 Appositions with profession nouns

Initial analysis of agreement occurrences with profession nouns showed, that most occurrences cannot be used in this study, because there is no possibility of determining whether the referent of the profession noun is male or female without the context of the surrounding text (sentences were analysed one by one). Because of this I wrote an additional script to extract information about appositions appearing with profession nouns, because if there is a feminine apposition to the profession noun, it is clear that the referent of the noun is female.[5]

The process of extracting information about agreement occurrences is almost identical to what was previously described, only here it applied exclusively to profession noun occurrences and had additional steps to find information about appositions. Using the script I first checked if the dependency head of the profession noun found in the corpus was an apposition. If that was the case, all of the dependents and the head of this apposition were treated as the dependents and the head of the profession noun and I looked for gender agreement between them and the profession noun. I calculated the distance again to make sure it is between them and the profession noun and not the apposition and I saved the word form and the morphological description of the apposition. Those fields were left empty if no apposition was found and the rest of the process was unchanged.

[5] The caveat to this approach is that it is not clear whether the feminine agreement with a profession noun is an occurrence of hybrid agreement, or if the agreement is not with the masculine profession noun but with the feminine apposition. This is addressed in the discussion section of this chapter.

2.1.3 Postprocessing of the agreement occurrences

The 214 917 initial observations were filtered to remove which was mistakenly included. There were 263 agreement observations with the gendered noun *pasztet*, which as a gendered noun means a girl or a woman whom the speaker finds unattractive, but it has another, much more common meaning: pâté, a kind of forcemeat. Most of the occurrences found in the corpus were in the second meaning, not useful for the study, therefore I removed this noun completely from the study.

Some of the occurrences found had to be parsing mistakes, such as large distances between target and controller, reaching up to 1124 intervening words, relative pronouns preceding the controller and targets of agreements being comprised only of digits, which cannot have any gender agreement marking. All of these were removed.

Another group of occurrences which could not be analysed were acronyms, which were marked as depreciative forms. I removed all of them, because analyzing acronyms is outside of the scope of this study. Similar with targets or controllers of agreement being plural, which could mean that they are a part of a coordination. Coordinate structures are a separate step of Corbett's Agreement Hierarchy and also outside of the scope of this study, so I removed them as well.

Some occurrences were suspicious, because they came from a sentence which had surprisingly many agreements (up to 8). Some of them were not actual sentences, but lists of people with certain functions, for instance a list of mayors chosen in local elections in an area. I looked through sentences with unusually many agreements and removed the ones which were actually lists, since any agreement there was a parsing mistake.

In some occurrences the controller was identified by the parser to have the wrong grammatical gender. This could result in inaccurate calculations of semantic agreement, therefore I also went through them to see what should be corrected and what should be discarded. This way some of the occurrences with the noun *zbieg* (meaning 'fugitive') were removed, because it was likely that the non-m1 occurrences of this noun were used in a different meaning, for example in the collocation *zbieg okoliczności* (meaning 'coincidence'). Similarly for the word *upiór* (meaning 'phantom'), which can also be used to mean a haunting memory, for *lump* (meaning 'tramp'), which can also mean old, distressed clothing, for *pajac* (meaning 'clown'), which can also mean a kind of doll and for *nurek* (meaning 'diver'), which can also refer to the act of diving.

The last part of filtering the data concerned the gender of the target of agreement. Possible values of gendered were predefined: m1 or any other with depreciative forms, m1 or f with profession nouns and with gendered nouns it depended on the lexeme. There were some occurrences for which the target of the gender was outside of the possible scope, which was most likely a result of parsing mistakes. For depreciative nouns there was no filtering done in this part, for profession nouns I assumed everything but the m1 or f agreements was parsing mistakes and removed those occurrences, and for gendered nouns I reviewed the incorrectly marked occurrences manually.

2.1.3.1 Additional processing of profession nouns

To figure out whether an occurrence of a profession noun referred to a man or a woman, I added information about appositions attached to the nouns. I removed occurrences with plural and non-F appositions and marked the ordering of the target, controller and apposition for each agreement, since it is possible that in some cases we agree the target with the gender of the apposition and not the gender of the controller. Only this selection of occurrences with profession nouns was saved for further analysis.

2.2 Results

2.2.1 Descriptive statistics

The first hypothesis this study aims to test is: there is a monotonous shift from one value of gender to another when going from attributive, through predicate to relative pronoun agreement. Additionally, the hypothesis proposed with the definition of generalized semantic agreement will be tested, which is that split hybrids (here the depreciative forms) are less likely to have generalized semantic agreement than lexical hybrids (here the socially gendered nouns and profession nouns). Before those hypotheses are addressed, we will look at the dataset extracted from the corpus.

After filtering the data which could not be used, there were 1722 observations, 443 of which were marked as semantic agreement. In Table 2.3 we can see the number of occurrences of agreement for all types of nouns (profession nouns, gendered nouns and depreciative forms) and for all types of targets (attributes, predicates and relative pronouns) and the percentages of semantic agreement for all those groups. The differeces between types of targets are also visualised in Figure 2.3.

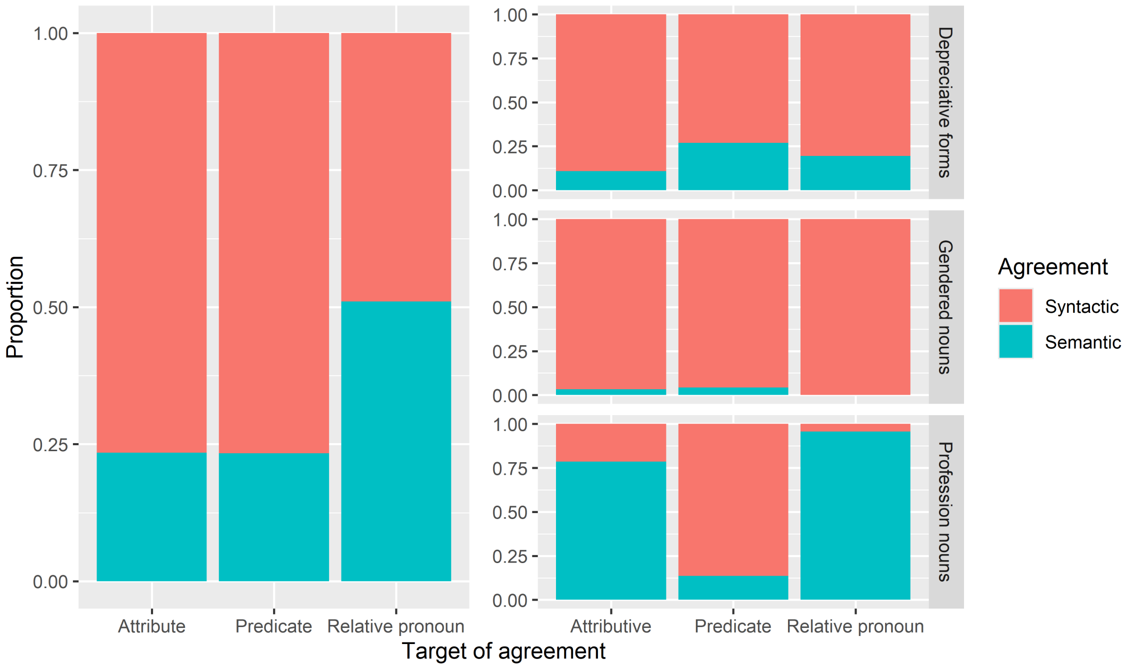


Figure 2.3: Proportions of semantic agreement for each type of target of agreement.

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
All nouns					
Attribute	891	209	23.46%	682	76.54%
Predicate	686	160	23.32%	526	76.68%
Relative pronoun	145	74	51.03%	71	48.97%
Total	1722	443	25.73%	1279	74.27%
Depreciative forms					
Attribute	343	37	10.79%	306	89.21%
Predicate	569	153	26.89%	416	73.11%
Relative pronoun	46	9	19.57%	37	80.43%
Total	958	199	20.77%	759	79.23%
Gendered nouns					
Attribute	343	11	3.21%	332	96.79%
Predicate	95	4	4.21%	91	95.79%
Relative pronoun	31	0	0%	31	100%
Total	469	15	3.2%	454	96.8%
Profession nouns					
Attribute	205	161	78.54%	44	21.46%
Predicate	22	3	13.64%	19	86.36%
Relative pronoun	68	65	95.59%	3	4.41%
Total	295	229	77.63%	66	22.37%

Table 2.3: Numbers and percentages of occurrences of agreement.

Out of 958 occurrences of agreement with depreciative forms, 20.77% had semantic agreement. The likelihood of semantic agreement was the lowest for attributes (10.79%) and highest for predicates (26.89%), leaving in between the relative pronouns (19.57%). For this type of controller there is no monotonous change of preference for semantic agreement from attributive to relative pronoun agreement, contrary to Corbett's prediction from the Agreement Hierarchy. In the following examples we can see depreciative forms with all targets of agreement in both possible forms, with the more likely form in bold. For all of them this was the form with syntactic agreement.

(22) a. **Tutej-sze**/tutej-si wykształciuchy nie są polskimi inteligentami.
local-F/local-M1 educated.people.M1 NEG are Polish intellectuals
"The local educated people are not the Polish intellectuals."

b. Mieszczuchy **wol-ały-by**/wol-eli-by, żeby chichotał i mrugał
city.dwellers.M1 **prefer-F-COND**/prefer-M1-COND that giggle and wink
porozumiewawczo.
knowingly
"The city dwellers would prefer him to giggle and wink knowingly."

c. Mnie najbardziej zbijają z nóg modnisie, **któ-re**/któ-rzy uważają się za
me most knock from legs dandys.M1 **which-F**/-M1 consider REFL as
orginalne.
original
"Dandys, who consider themselves original, surprise me the most."

According to one of the hypotheses, semantic agreement should be more acceptable with lexical hybrids than with split hybrids, which would mean that socially gendered nouns would have a higher percentage of semantic agreements than depreciative forms. This is not reflected in the collected corpus data, where only 3.2% of occurrences with gendered nouns have semantic agreement. Here the percentages of semantic agreement change monotonously from attributive to relative pronoun agreement, although it is the syntactic agreement that becomes more likely as we go along the Agreement Hierarchy. Considering this, the data from socially gendered nouns fits only the simple constraint of the Agreement Hierarchy (Corbett, 2023). The following examples come from the corpus and show which of the possible agreements are preferred.

(23) a. Zachichotał jak **stuprocentow-a**/stuprocentow-y ciota.
giggled like **hundred.percent-F**/hundred.percent-M1 faggot.F
"He giggled like a 100% faggot."

b. Kobiecisko **przytulił-o**/przytulił-a się histerycznie do mnie.
woman.N **hugged-N**/hugged-F REFL histerically to me
"The woman hugged me histerically."

c. Pamiętasz historię dziewczęcia, **któr-e**/któr-a uciekło z balu?
remember story girl.N **which-N**/which-F ran.away from dance
"Do you remember the story of the girl, who ran away from the dance?"

There were 295 occurrences of agreement with profession nouns, 77.63% of which had semantic agreement. This percentage is larger than for other types of nouns, however this is most likely because all of the profession noun occurrences included in the analysis appeared with feminine appositions, which increase the likelihood of semantic agreement. I will explore this more later when considering the order in which the apposition, controller and target appear in the sentences. With profession nouns it was more likely that semantic agreement would be used in attributes (78.54% of all attributive agreement occurrences) and in relative pronouns (95.59% of all relative pronoun agreement occurrences), but not in predicates (13.64% of all predicate agreement occurrences). This, again, does not follow the simple constraint of the Agreement Hierarchy proposed by Corbett (2023), according to which there should be a monotonic change from attributive to relative pronoun agreement. The preferences found in the corpus can be seen in the following examples.

(24) a. Większość złożyli w końcówce kadencji obecne-go/**obecne-j** burmistrz
majority submitted in ending term current-M1/**current-F** mayor.M1
Stefanii Sulińskiej.
Stefania Sulińska

“The submitted the majority of them at the end of the term of the current mayor, Stefania Sulińska.”

b. One są gotowe do adaptacji do roku 2008 – **zapewniał-∅**/zapewniał-a
they are ready for adaptation to year 2008 - **assured-M1**/assured-F
minister edukacji Krystyna Łybacka.
minister.M1 education Krystyna Łybacka

“They are ready to be adapted before the year 2008 - assured the minister of education Krystyna Łybacka.”

c. Felicity Huffman gra panią naukowiec, któr-y/**któr-a** w arktycznej
Felicity Huffman plays lady scientist.M1 which-M1/**which-F** in arctic
stacji zostaje zarażona pasożytem z kosmosu.
station becomes infected parasite from space

“Felicity Huffman plays the scientist, who is infected with a parasite from space in the arctic station.”

It is possible that agreement with profession nouns is more likely to be semantic than with other types of nouns, because in those occurrences there were also feminine appositions to the nouns. In those situations it is possible that speakers chose feminine agreement on targets (which for profession nouns is semantic agreement), because they were actually agreeing the gender of the target with the gender of the apposition. To have more insight into the structure of those sentences I marked the order in which the appositions (A), controllers (C) and targets (T) appeared with each occurrence of agreement. Figure 2.4 shows the proportions of semantic to syntactic agreement in occurrences with each possible order of appositions, controllers and targets.

Figure 2.4: Proportion of syntactic and semantic agreements with ordering of the controller, target and apposition

Three of the orders shown in the figure noticeably favor semantic agreement over syntactic: apposition-controller-target, apposition-target-controller and target-apposition-controller. In contrast to the other orders, for these the apposition appears before the controller. Since this seems to be the most important information coming from the order variable, for statistical modeling instead of accounting for the order we will account for whether the apposition appears first (before the controller) or not.

Seeing that, when the apposition appears before the controller, the agreement is almost always semantic, we can look at how the proportions of semantic agreement change if we assume that we should exclude occurrences with apposition first, because the speakers agree the gender of the target with the gender of apposition and not the semantic gender of the controller. Figure 2.5 is identical to Figure 2.3, only with occurrences of profession nouns with appositions first excluded.

Target of agreement	N	Semantic agreement		Syntactic agreement	
		N	%	N	%
Attribute	46	3	6.52%	43	93.48%
Predicate	19	3	15.79%	16	84.21%
Relative pronoun	6	3	50%	3	50%
Total	71	9	12.68%	62	87.32%

Table 2.4: Numbers and percentages of occurrences of agreement with profession nouns without occurrences with appositions first.

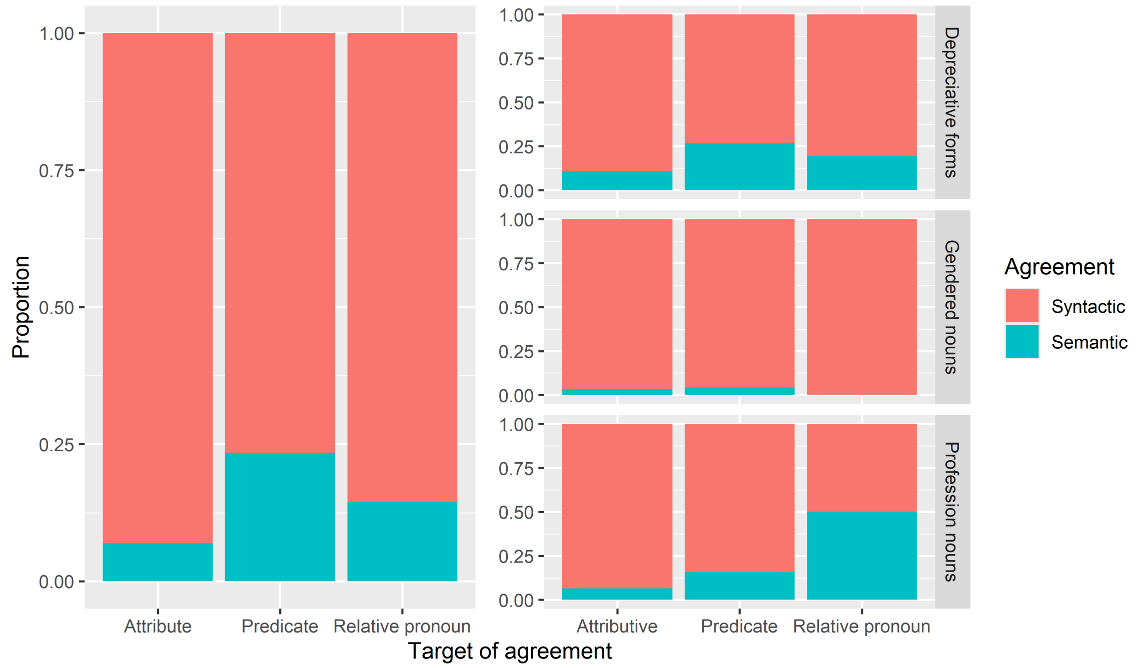


Figure 2.5: Proportions of semantic agreement for each type of target of agreement without occurrences of profession nouns with preceding appositions.

We can see that the overall proportions of semantic agreement (left side of the figure) are now unambiguously not monotonic, and therefore not compatible with any version of the Agreement Hierarchy constraints. The proportions for the profession nouns only, however, now show a monotonic shift from preference for almost exclusively syntactic agreement with attributes, to close to 50% syntactic and semantic agreement.

2.2.2 Statistical models

For statistical modeling of the corpus data I used the `glmer` function from the `lme4` R package and the `glm` function from the `stats` package. The first was a binomial generalized linear mixed model, because the outcome was binary: whether semantic agreement was chosen or not. The model used all of the 1722 observations. Fixed effects were the type of agreement target (attribute, predicate or relative pronoun), distance between the target and controller, target placement (preceding or following the controller), type of controller (gendered noun, profession noun or depreciative form) and apposition placement (preceding or following the controller). The corpus contains texts from 13 different types of sources – this was included as the random effect in the model. The full list of the model's coefficients is presented at the end of this chapter.

Types of targets and the distance were not significant as predictors of semantic agreement. Target being placed after the controller increased the likelihood of semantic agreement ($p<0.001$), as well as placing an apposition before the controller ($p<0.001$). Among the noun types the depreciative forms were the baseline, because the hypothesis was that as split hybrids semantic agreement would be less likely to occurr with them than with the lexical hybrids: gendered and profession nouns. Compared to depreciative forms then, socially gendered nouns turned out to be less likely to have semantic agreement ($p<0.001$), while profession noun were more likely to have semantic agreement ($p<0.001$).

The second model was a binomial generalized linear model and used only the occurrences with depreciative forms. The independent variables were the type of agreement target, distance and target placement. Out of these, the only significant effect was that of the target placement – target following the controller increased the likelihood of semantic agreement ($p<0.001$).

I also computed a binomial generalized linear model using only the occurrences with socially gendered nouns, with the same independent variables as for the previous model: the type of agreement target, distance and target placement. Here the only significant predictor was the distance – larger distance increased the likelihood of semantic agreement with gendered nouns ($p<0.05$).

Finally, there was also a binomial generalized linear model which used only the occurrences with profession nouns. In this model, additionally to the predictors in the previous one, there was the placement of the apposition. Distance and target placement were not significant in this model. Contrary to what the Agreement Hierarchy indicates, the odds of attributive agreement being semantic were approximately 16.16 times higher than the odds of predicate agreement being semantic ($p<0.05$). For relative pronouns the odds were 25.84 times higher than for predicates ($p<0.05$). There was also a significant effect of apposition placement – the apposition appearing before the controller increased the likelihood of semantic agreement ($p<0.001$).

```
Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: semantic_agr ~ type_of_agreement + distance + target_first +
    source_cat + appos_first + (1 | type_of_text)
   Data: .

     AIC      BIC   logLik -2*log(L)  df.resid
  1161.9   1210.9   -571.9    1143.9      1713

Scaled residuals:
     Min       1Q   Median       3Q      Max
-11.7801  -0.4704  -0.2502   0.0808   8.5388

Random effects:
 Groups       Name        Variance Std.Dev.
 type_of_text (Intercept) 0.06498  0.2549
Number of obs: 1722, groups:  type_of_text, 13

Fixed effects:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                  -5.99309    0.49265 -12.165  < 2e-16 ***
type_of_agreementattributive -0.36675    0.22379  -1.639    0.101
type_of_agreementrelpron     -0.34526    0.33058  -1.044    0.296
distance                      0.02303    0.03811   0.604    0.546
target_first0                 0.97644    0.21494   4.543 5.55e-06 ***
source_catgendered           -1.90220    0.28967  -6.567 5.14e-11 ***
source_catprofession          1.81877    0.32474   5.601 2.14e-08 ***
appos_first                   4.22701    0.42209  10.015  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) typ_f_grmntt typ_f_grmntr distnc trgt_0 src_ctg src_ctp
typ_f_grmntt -0.272
typ_f_grmntr -0.008  0.125
distance     -0.047  0.174        0.104
targt_frst0  -0.443  0.553       -0.093       -0.067
src_ctgndrd  -0.047 -0.148       -0.099       -0.028 -0.005
src_ctprfss  -0.058 -0.168       -0.117       -0.101  0.061  0.109
appos_first  -0.862 -0.055       -0.032       -0.058  0.085  0.010    0.016
```

Figure 2.6: Model with all types of nouns

```
Call:
glm(formula = semantic_agr ~ type_of_agreement + distance + target_first,
    family = binomial(link = "logit"), data = .)

Coefficients:
                             Estimate Std. Error z value Pr(>|z|)    
(Intercept)                  -1.85889    0.23869  -7.788 6.82e-15 ***
type_of_agreementattributive -0.47954    0.24973  -1.920   0.0548 .  
type_of_agreementrelpron     -0.61260    0.38700  -1.583   0.1134    
distance                     -0.03910    0.04125  -0.948   0.3432    
target_first0                 1.07946    0.24116   4.476 7.60e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 978.93  on 957  degrees of freedom
Residual deviance: 920.38  on 953  degrees of freedom
AIC: 930.38

Number of Fisher Scoring iterations: 4
```

Figure 2.7: Model with depreciative forms

```
Call:
glm(formula = semantic_agr ~ type_of_agreement + distance + target_first,
    family = binomial(link = "logit"), data = .)

Coefficients:
                              Estimate Std. Error z value Pr(>|z|)    
(Intercept)                    -4.6244     0.9319  -4.962 6.97e-07 ***
type_of_agreementattributive    0.8582     0.8527   1.006 0.314223    
type_of_agreementrelpron      -14.7725  1124.8988  -0.013 0.989522    
distance                        0.4417     0.1220   3.619 0.000295 ***
target_first0                   0.2693     0.6813   0.395 0.692599    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 132.79  on 468  degrees of freedom
Residual deviance: 118.40  on 464  degrees of freedom
AIC: 128.4

Number of Fisher Scoring iterations: 17
```

Figure 2.8: Model with gendered nouns

```
Call:
glm(formula = semantic_agr ~ type_of_agreement + distance + target_first +
    appos_first, family = binomial(link = "logit"), data = .)

Coefficients:
                            Estimate Std. Error z value Pr(>|z|)
(Intercept)                  -5.1604     1.4187  -3.638 0.000275 ***
type_of_agreementattributive  2.7827     1.2596   2.209 0.027158 *
type_of_agreementrelpron      3.2520     1.3174   2.469 0.013566 *
distance                      0.3212     0.2448   1.312 0.189503
target_first0                 0.9457     0.9044   1.046 0.295722
appos_first1                  5.7027     0.7131   7.997 1.27e-15 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 313.636  on 294  degrees of freedom
Residual deviance:  80.749  on 289  degrees of freedom
AIC: 92.749

Number of Fisher Scoring iterations: 7
```

Figure 2.9: Model with profession nouns

2.3 Discussion

Chapter 3

Experimental study

3.1 Methodology

3.1.1 Procedure

There were two acceptability judgement experiments conducted as a part of the study: one using stimuli with profession nouns and the other with depreciative forms. Both experiments had the same design, which was 3x2 within-subject. The independent variables were the agreement options (semantic or syntactic) and type of agreement with ordering of target and controller: predicate with target first (prenominal), predicate with controller first (postnominal) and relative pronoun (always with controller first, since it is impossible to place the relative pronoun before its antecedent). The dependent variable was the rating of acceptability of given sentences.

The experiment was conducted in Polish. It was recommended to participants to complete the experiment using a computer or tablet, to make sure the instructions and stimuli were displayed correctly. On the first page participants were told their task was to judge whether the displayed sentences sounded naturally or not. They were also informed that participation in the study was voluntary and they could withdraw at any time. To proceed to the study they had to tick a box to confirm they consent. Next, they saw the instructions page, which had the following information.

Your task will be to judge how naturally the displayed sentences sound. If you aren't sure what this means, consider whether you could hear or say the given sentence. You will judge the sentences on a 5-point scale, from "completely unnatural" to "completely natural". You will see a control question after each sentence.

Before we proceed to the study, you will see 3 practice questions that will allow you to familiarize yourself with the task. After the practice session you will proceed to the study, which consists of 60 questions. During the study you cannot go back to the previous questions and there is no time limit to answer the questions, but answer intuitively, without thinking too much.

Out of 60 trials, 24 were with experimental items. They were arranged using the Latin square design – in 6 lists, so that each item appears on a list once and each list has 4 items per condition. Each participant was assigned a list randomly and the order of items was randomized each time. Each trial was followed by a simple comprehension question, to ensure that participants are paying attention to the experiment.

In the experiment on profession nouns the instruction in each trial was "Read the sentences below and judge how naturally the ***second*** one of them sounds", for the experiment on depreciative forms it was "Read the sentence below and judge how naturally it sounds". As was said in the instructions for participants, judgements were given on a 5-point scale. Such a scale is familiar for Polish speakers, because it is similar to the grading scale in Polish schools (1-6 with 6 being the best) and in Polish universities (2-5 with 5 being the best). The scale was labelled only on the ends with *całkowicie nienaturalne* (meaning 'completely unnatural') and *całowicie naturalne* (meaning 'completely natural').

After participants gave judgements on 60 sentences, there was a short demographic survey asking about their age, gender, first language(s) and the country or countries where they were raised. All of the response fields for those questions allowed for any kind of text answer to be typed. The duration of the whole experiment was 15-20 minutes.

3.1.2 Materials

Experimental items were constructed so that the agreement controller was either in the first or the final position in the sentence, the agreement target was immediately adjacent to the controller and there were no other words in the sentence that agreed with the controller in gender. To make sure that the last constraint was not violated, all of the sentences with relative pronoun agreement had to be in present tense (because predicates in present tense do not agree with the subject in gender), while the sentences in other conditions were in past tense. The following subsections describe how the nouns for the experimental items were selected.

3.1.2.1 Experimental items with depreciative forms

Depreciative forms appear in the paradigms of m1 nouns and morphologically they resemble nouns of genders other than m1. In Corbett's approach then, when using those nouns in a sentence we have a syntactic reason to use non-m1 agreement and a semantic reason to use m1 agreement. To make sure that semantic reason is clearly pointing to m1, the nouns selected for this experiment had to be commonly perceived as masculine. To ensure that, the selection process relied on a norming study conducted by Misersky et al. (2014), where they collected information about gender stereotypicality for role nouns in multiple Eurpean languages.

There were 22 nouns that fulfilled the following criteria:

1. Perceived to be at least 60% masculine according to the norming study.

2. Appear at least a 1000 times in the balanced version of the National Corpus of Polish.

3. Depreciative form is not syncretic with the non-depreciative.

4. Does not have disturbing, violent connotations (e.g. rapist, murderer)

5. Depreciative form is not syncretic with a different noun (e.g. noun *technik* (meaning 'technician') is *techniki* in depreciative, which has the same shape as the word for 'techniques' and could lead to confusion)

6. Is not a nominalized adjective (for which depreciative forms are syncretic with the feminine versions of these nouns)

Additionally, there were two nouns that did not occur in the norming study: *proboszcz* (meaning 'parish priest') and *chłopak* (meaning 'boy'). The former fulfills all of the criteria above, besides being perceived as at least 60% masculine in the norming study, since it is not included there. However, it can be included in the experiment because it is a noun commonly used in Polish and, as a subtype of a priest, it counts as being perceived as masculine.

As Skowrońska (2011) reports, the latter of the two additional nouns is problematic for Polish speakers. In a section dedicated specifically to the lexeme 'chłopak' she reports that some Polish speakers are taught that the non-depreciative form of the noun is incorrect. Because of this, hybrid agreement is very commonly used with this particular noun. The author also looked through the National Corpus of Polish to see a few examples of the noun with attributive and predicate agreement and she finds that predicates with this noun are mostly seen with semantic agreement, while attributes occur almost exclusively with syntactic agreement. This is in line with Corbett's predictions and the current study can show whether this tendency extends to the further points of the Agreement Hierarchy.

All of the nouns chosen for this experiment can be found in the appendix. In (25) there is an example of an experimental item with one of the nouns in depreciative form.

(25) Example of an item in the experiment with depreciative forms in all conditions and with its comprehension question.

a. f/m1, prenominal

Na	nagrania	do	Ameryki	Południowej	poleciały/polecieli	reżysery.
for	shooting	in	America	South	flew:F/flew:M1	directors(M1)

The directors flew to South America for shooting.

b. f/m1, postnominal

Reżysery	poleciały/polecieli	na	nagrania	do	Ameryki	Południowej.
directors(M1)	flew:F/flew:M1	for	shooting	to	America	South

The directors flew to South America for shooting.

c. f/m1, relative pronoun

Reżysery, które/którzy lecą na nagrania do Ameryki Południowej,
directors(M1) who:F/who:M1 are.flying for shooting to America South
nie odpowiadają teraz na maile.
NEG responding now to emails

Directors, who are flying to South America to shoot, are not responding to emails right now.

d. comprehension question
Czy w zdaniu wspomniano o Azji?
Did the sentence mention Asia?

3.1.2.2 Experimental items with profession nouns

The process of selecting nouns for this experiment was similar to that for the corpus study. The study by Szpyra-Kozłowska (2019) was used to find masculine nouns, for which it is difficult to propose a feminine version. Nouns were arranged from the most times participants of the study refused to give a feminine version of the noun, to the least. This formed a ranking, from which the following nouns were excluded:

1. *ksiądz* (meaning 'priest'), *pastor* (meaning 'pastor') and *proboszcz* (meaning 'parish priest'), because in Poland the population of female priests is almost nonexistent
2. offensive words (e.g. *dureń* (meaning 'fool'))
3. words with less than a 1000 occurrences in the balanced version of the National Corpus of Polish (with prose excluded to make sure the words are used in everyday language)

Here each experimental item also required a sentence of context preceding the target sentence. The nouns used in this experiments are masculine and can refer to both men and women, so context is needed for participants to know that in this situation the noun refers to a woman. All of the items for this experiment can be found in the appendix and an example of an item can be found in (26).

(26) Example of an item in the experiment with profession nouns in all conditions, with its context and the comprehension question.

a. context
W tym tygodniu odwiedzają nas mamy uczniów z naszej klasy, żeby opowiedzieć o swojej pracy.
Moms of kids from our class are visiting us at school this week to talk about their careers.

b. f/m1, prenominal

We wtorek przyszła/przyszedł architekt.
on Tuesday came:F/came:M1 architect(M1)

The architect came on Tuesday.

c. f/m1, postnominal

Architekt przyszła/przyszedł we wtorek.
architect(M1) came:F/came:M1 on Tuesday

The architect came on Tuesday.

d. f/m1, relative pronoun

Architekt, która/który przyjdzie we wtorek, należy do Rady
architect(M1) who:F/who:M1 will.come on Tuesday belongs to council
Rodziców.
of.parents

The architect who will come on Tuesday is a member of the Parents' Council.

e. comprehension question
Czy dzień tygodnia, który był wspomniany w zdaniu, to czwartek?
Was Thursday the day of the week mentioned in the sentence?

3.1.2.3 Other items

Both experiments also had 12 control items, half correct, to establish a threshold of acceptable and unacceptable sentences. The incorrect sentences contained an attraction error with the subject, the correct ones were the same but without the error. Half of the controls had human subjects and half had non-human, this was distributed evenly within the conditions. In the profession nouns experiment these items were preceded by a sentence of context, so that the experimental item would not be too easy to distinguish. An example of a control item can be found in (27).

(27) Example of a control item in both conditions and the context sentence.

a. context
Nie wszystkie dzieci chciałyby mieć wieczne wakacje.
Not all kids want to be on holidays forever.

b. correct/incorrect

Syn sąsiadki jest szczęśliwy/szczęśliwa że idzie do szkoły.
son(M1) of.neighbor(F) is happy:M1/happy:F that goes to school

The son of the neighbor is happy to go to school.

To make sure the main focus of the study is not too obvious to participants, there were also 24 filler items. As with control items there was a context sentence preceding the fillers in the profession noun experiment. An example can be found in (28).

(28) Example of a filler item with its context.

a. context
Szukam zeszytu w którym babcia zapisywała hasła.
I'm looking for the notebook where grandma wrote down her passcodes.

b. Znalazłem na strychu sejf i nie wiem jak go otworzyć.
I.found on attic strongbox and NEG know how it open
I found a strongbox in the attic and I don't know how to open it.

3.1.3 Participants

Participants for both studies were recruited using Prolific. The target population on the platform was filtered to only include people who currently live in Poland, have Polish nationality and speak Polish as their first language. [PUT HERE age range and median, genders, if theres anything about first languages and countries, exclusion criteria that might come up] Participants were paid 3.20 GBP (3.7 EUR) for participation. This is 15.69 PLN for 20 minutes which corresponds to 47.07 PLN hourly, so around 150% of minimum hourly wage in Poland as of 2026.

3.1.4 Hypotheses

Based on the corpus study, one hypothesis is that syntactic agreement is preferred over semantic, so items with semantic agreement should have an overall lower acceptability rating than the items with syntactic agreement. The semantic agreement should, however, be rated higher than the ungrammatical control sentences.

Additionally, according to the stronger constraint of Corbett's Agreement Hierarchy, the acceptability of semantic agreement should increase as we go rightwards along the hierarchy. Here this means that semantic predicate agreement has a lower average acceptability rating than semantic relative pronoun agreement. Another hypothesis concerns the Hierarchy of Agreement Sources, based on which semantic agreement with split hybrids is less acceptable than with lexical hybrids. For this study the split hybrids are depreciative forms and lexical hybrids are profession nouns, therefore semantic agreement in the the profession nouns experiment should have an overall higher acceptability than semantic agreement in the depreciatives experiment. Finally, according to Corbett the target of agreement preceding the controller decreases the likelihood of semantic agreement, so in the experiments the prenominal condition should have a lower acceptability than the postnomial condition.

3.2 Results

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